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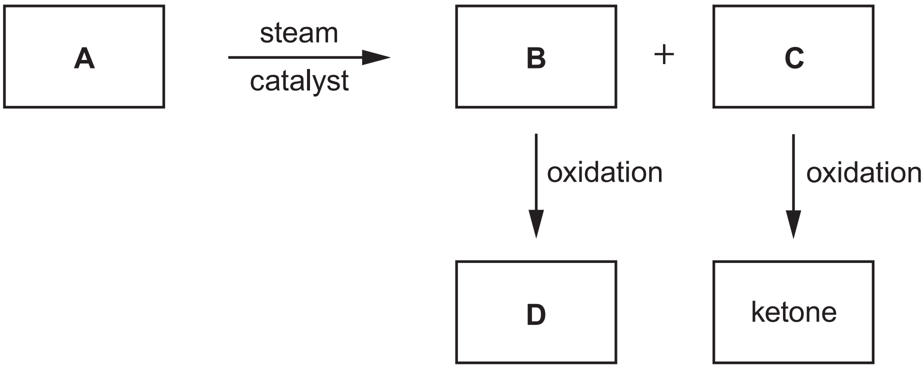
9. Look at the reaction scheme below and other information about compounds **A** to **D**.



A is a straight chain hydrocarbon. It contains 85.7 % carbon by mass.

B and **C** are structural isomers.

D is soluble in water. 0.93 g of **D** is dissolved in water to make 100 cm^3 of solution. 25.0 cm^3 of this solution is titrated against 0.100 mol dm^{-3} aqueous sodium hydroxide. 26.40 cm^3 of sodium hydroxide is needed for neutralisation. **D** reacts with sodium hydroxide in a ratio of 1:1.

(a) Calculate the empirical formula of **A**. [2]

Empirical formula ....

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(b) Show that the relative molecular mass (M_r) of **D** is 88. [3]

(c) Give the names of compounds **A** to **D**. [4]

A ....

B ....

C ....

D ....

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(d) Name a suitable reagent for the conversion of **B** to **D**. [1]

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